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Supplementary information

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Glacier-preserved Tibetan Plateau viral community probably linked to warm–cold climate variations

In the format provided by the authors and unedited

Supplementary Discussion:

Assembly pipeline optimization

We compared 18 assembly pipelines in terms of the number of assembled contigs and putative viral contigs (both ≥5 kb) (See Methods; Extended Data Fig. 1). These comparisons found that a combination of “single-cellSPAdes + -k 21,33,55,77,99,127” generated significantly higher number of both assembled and viral contigs than other pipelines, including assemblies by the tool MEGAHIT (Extended Data Fig. 1). Within the above combination (i.e., single-cellSPAdes + -k 21,33,55,77,99,127), the assemblies benefited from read error correction; the read deduplication did not obviously impact the assemblies, while it generated more assembled contigs in all the three tested metagenomes and more viral contigs in two of the three tested metagenomes when a combination of “read error correction + single-cellSPAdes + -k 21,33,55,77,99,127” was used (Extended Data Fig. 1). Taken together, we recommend using a combination of “read deduplication + read error correction + single-cellSPAdes + -k 21,33,55,77,99,127” for assembling the low-biomass glacier metagenomes and the subsequent viral identification, and we used this optimized pipeline to assemble all glacier metagenomes in this study, generating 1,705 vOTUs (≥5 kb; hereafter refers to Guliya vOTUs or Guliya viruses). Our recommendation for assembly method agrees with the previously optimized assembly pipeline for PCR-amplified low-biomass metagenomes by assessing only the assembled contigs ≥10 kb[1], while here we extended the evaluation to the improvements of generating shorter assembled contigs and viral contigs (both ≥5 kb).

Viruses infect dominant microbes

The microbial communities of the nine ice samples studied were dominated by three phyla (relative abundance >1% in all nine samples; their summed relative abundance comprised 97.5–99.6% of each of the nine communities): Proteobacteria, Bacteroidetes, and Actinobacteria. Guliya viruses infecting the above three dominant phyla were also abundant (relative abundance >1%) in all nine samples (Supplementary Fig. S3A). These results suggest that viruses may have important impacts on glacier surface ecosystems via infecting dominant microbial phyla over >41,000 years. To further investigate the potential influences of viruses on host ecology, we calculated lineage-specific virus-host ratios (VHRs) to assess how viral infection dynamics for each of the three dominant phyla (Proteobacteria, Bacteroidetes, and Actinobacteria) varied. The VHRs of these phyla varied over time, with the ratio of Actinobacteria persistently higher than that of Bacteroidetes and Proteobacteria (Supplementary Fig. S3B), reflecting potentially higher viral infection pressure on Actinobacteria over time in the ancient ecosystems.

At the genus level, we also found that abundant viruses infected *Cryobacterium* (average and range of relative abundances: 13.9% and 2.5–24.6%, respectively; within Actinobacteria), *Flavobacterium* (14.7% and 5.4–26.9%; within Bacteroidetes), and *Polaromonas* (10.4% and 6.3–12.8%; within Proteobacteria), which were the three abundant microbial lineages across all the nine ice samples (Fig. 3A). Analyses of VHRs found that the ratio of *Flavobacterium* was persistently higher than that of *Cryobacterium* and *Polaromonas* (Fig. 3B), reflecting potentially higher viral infection pressure on *Flavobacterium* than *Cryobacterium* and *Polaromonas* over the entire record in the ancient ecosystems. *Cryobacterium*, *Flavobacterium*, and *Polaromonas* are three commonly dominant genera in glacier-preserved communities and on the modern glacier surfaces with many members having cold-adapted genomic features to enable them to thrive under cold conditions and potentially participating in carbon cycling[2-7]. The relatively high abundance of these genera and their associated viruses suggests that the recovered viruses infected abundant microbial groups and thus might play a major role in this extreme ecosystem by influencing their hosts when they were active prior to freezing. Together, these results used glacier-archived DNA to reveal the ancient functioning of viruses and indicate that glacier viruses may have important impact on ecosystems via infecting dominant microbes.

Potential high infection pressure of Patescibacteria

High viral infection pressure can occur even in the case of rare microbial lineages. To investigate this, we expanded the examination of VHRs to include the top eight abundant phyla: Proteobacteria, Actinobacteria, Bacteroidetes, Patescibacteria, Cyanobacteria, Chloroflexi, Firmicutes, and Verrucomicrobia, by decreasing in their relative abundances. Notably, the VHR of Patescibacteria was significantly higher (average ~20 times higher; $p=0.007$) than that of the other seven phyla (Supplementary Fig. S4A), indicating potentially higher viral infection pressure on Patescibacteria. In support of this finding, we examined the genomes of the 2,358 TP glacier MAGs and found that the genome size, GC content, percentage of genomes containing a CRISPR, and the number of spacers in each CRISPR (reflecting phage defense capacity) were significantly lower in Patescibacteria than the other seven phyla (Supplementary Fig.

S4B–E) – in accord with the findings of Patescibacteria MAGs from the oligotrophic groundwater[8]. Therefore, high viral infection pressure on Patescibacteria could be due to less CRISPRs and spacers on their genomes. Our findings were also consistent with a previous report that Patescibacteria is one of the most infected lineages in Arctic peat soil – an ecosystem characterized by sub-freezing temperature and low nutrient availability[9].

Patescibacteria has recently received increasing interest because this microbial group expanded our view of the tree of life[10]. In addition to the unique genomic features described above, Patescibacteria has been characterized by small cell size (thus higher ratio of surface area that may benefit substance exchange in low-nutrient environments), reduced non-essential metabolic capabilities, and has the potential for co-metabolism interdependencies using a symbiotic lifestyle with other microbes[8,11]. Owing to these lifestyle capabilities, Patescibacteria have been repeatedly found in cold and/or oligotrophic ecosystems[8,12]. In glacier ecosystems however, Patescibacteria's adaptive traits and ecological roles have not been investigated, while a large number of Patescibacteria MAGs (n=79) have recently been identified in snow, cryoconite, and young ice from the surfaces of eight TP glaciers[7], indicating their prevalence and potentially important ecological roles in this extreme cold and oligotrophic environment. Thus, our results highlighted that in the glacier environment Patescibacteria is potentially an understudied but ecologically important microbial group, which likely has a high infection pressure from glacier viruses over tens of thousands of years.

Viruses can potentially impact hosts' biosynthesis of vitamin B_{12} and sulfur metabolism

Among the AMGs involved in the "metabolism of cofactors and vitamins" was the cobaltochelatase gene (*cobT*) that was identified from a Guliya vOTU with a complete genome (with 55-bp repeats at the genomic ends; Fig. 3D) and has not been characterized in viruses before. This gene encodes a component of cobaltochelatase, which catalyzes cobalt insertion in the corrin ring during the biosynthesis of vitamin B_{12}[13]. Only a small fraction of microbes can synthesize B_{12}, which, however, is an important compound needed by almost all microbes for cellular metabolisms (e.g., DNA production and protein synthesis) and growth and can impact cell behavior including stress resistance (e.g., irradiance and oxidative stresses) to further shape microbial community structure, stability, and overall function[14-16]. On glacier surfaces under extreme conditions (low temperature and nutrient availability, strong radiation and oxidative stress), B_{12}, as a member of cofactors and vitamins category, could potentially help microbes cope with the stressful conditions[7]. Phylogenetic analysis revealed that this virus-encoded *cobT* gene (*vCobT*) might have been transferred from microbes of the genus *Ralstonia* (Supplementary Fig. S6A), some members of which are able to participate in vitamin B_{12} metabolism[17]. Evaluating the protein sequence and amino acid changes indicated that vCobT is likely functional. First, it contained the conserved MIDAS motif (Metal Ion-Dependent Adhesion Site; DXSXS) (Supplementary Fig. S7A), which provides the binding site of a single cation and is crucial for the protein function[18]. Further, the number of nonsynonymous mutations relative to the number of synonymous mutations (*pN/pS*), a measure of selection pressure, was determined by recruiting metagenomic reads from ice samples to this *vCobT* and then calculating *pN*/*pS*. This returned a value of 0, which indicated that the CobT was likely functional and under purification selection. This result was supported by analyzing the evolutionary dynamics of *cobT* homologs across lineages (see Methods) which implies that the *cobT* was under purification selection (average *dN*/*dS* = 0.106) and likely remained functional (Supplementary Data 10).

Microbes are important participants in biogeochemical cycles of carbon, nitrogen, and sulfur in the environments including supraglacial ecosystems[19-21]. We discovered a glacier virus that potentially impact hosts' sulfur metabolism by encoding a phosphoadenosine phosphosulfate reductase gene (*cysH*; Fig. 3E), which encodes a critical enzyme for assimilatory sulfate reduction, and has been frequently identified in environmental viruses (e.g., Ref. [22,23]) but has not been reported in glacier viruses. Sulfur is an essential element of microbes, and its reduction process is critical for microbial activities including respiration and energy generation[24]. We propose that glacier viruses are potentially able to impact microbial activities via modulating their sulfur metabolisms and further influence the biogeochemical sulfur cycles in the glacier environment. This virus-encoded *cysH* gene (*vCysH*) might have been transferred from microbes of the genus *Janthinobacterium*, a very common glacier lineage with cold-adaptive molecular capability[5,25], to the virus (Supplementary Fig. S6B). Further exploration of the conserved amino-acid motifs and evolutionary dynamics within species and across lineages indicated that the vCysH is likely functional and under purification selection (*pN/pS* = 0; average *dN/dS* = 0.063) (Supplementary Fig. S7B & Data 11). Overall, we propose that the viruses can potentially encode AMGs to functionally impact the glacier microbes and community complex, including the impacts on stress resistances via synthesizing vitamin B_{12} and on sulfur metabolism via reducing sulfate.

Supplementary Figures:

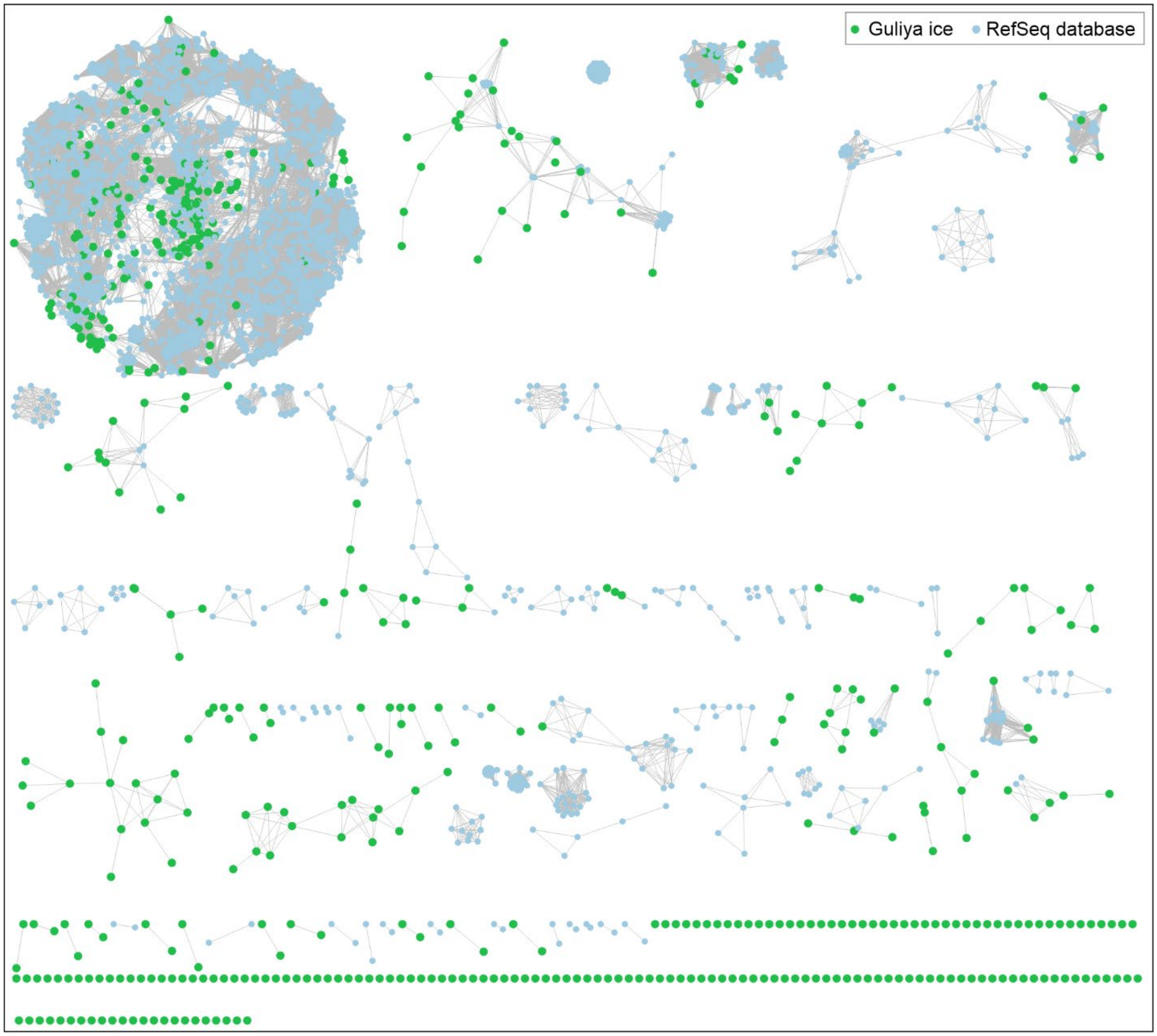


Figure S1. Network clusters of viruses from this study (in green) and NCBI RefSeq database (light blue). Each node represents one viral genome/contig; the edge between nodes represents a significant relationship between the two connected viral genomes/contigs with the shorter length accounting for stronger connection strength. The details of viral clusters and statistical results are provided in Supplementary Data 3.

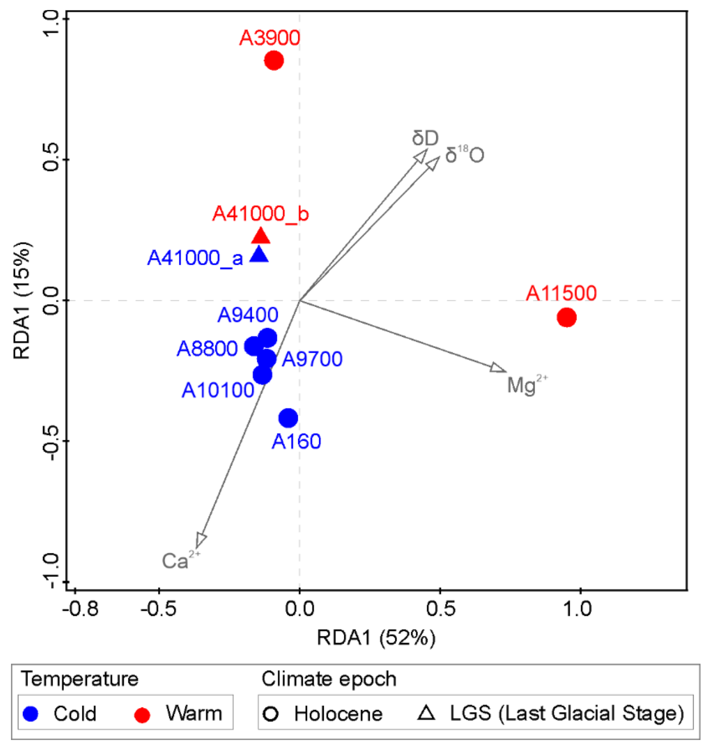


Figure S2. Ecological drivers of viral communities by dbRDA analysis. None of the tested factors had a significant impact ($p<0.05$) on the Guliya viral communities, as measured by Monte Carlo permutation tests. The four variables Mg^{2+}, $\delta^{18}O$, δD, and Ca^{2+} were selected to illustrate the dbRDA plot as they had the top four impacts, though not statistically significant, with p values 0.07, 0.108, 0.166, and 0.190, respectively (two-sided, false discovery rate correction; see Supplementary Data 5). Samples from cold and warm conditions are shown in blue and red, respectively; samples from the two major climate epochs Holocene and Last Glacial State (LGS) are indicated by circle and triangle shapes, respectively. The sample A11500, which was collected at the beginning of the Holocene, contained the most distinct community compared to the other samples, as also indicated by the PCoA plot in Fig. 2C.

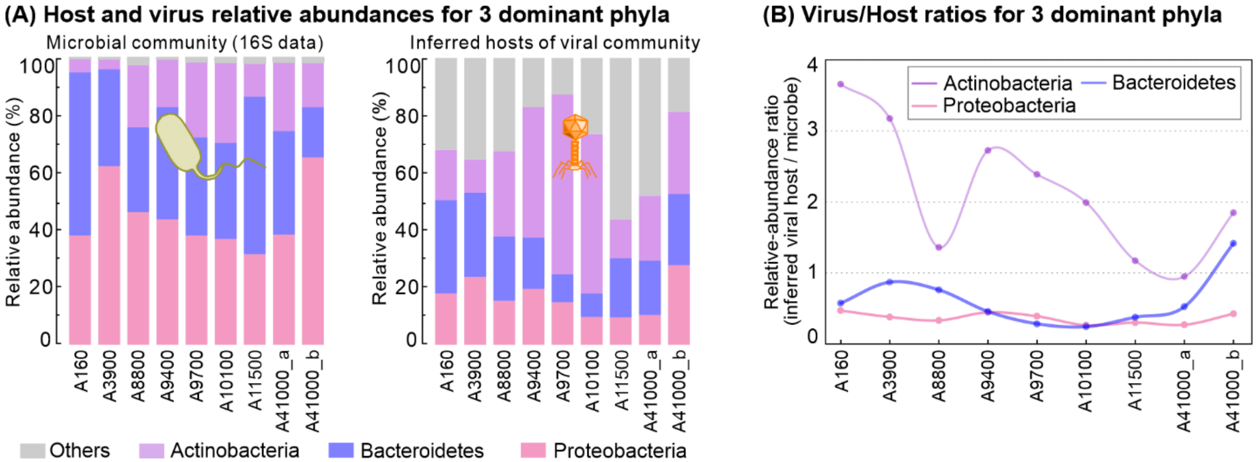


Figure S3. Historical changes of virus infections of hosts at phylum level. (A) Relative abundances of the three most dominant phyla (Actinobacteria, Bacteroidetes, and Proteobacteria; via 16S rRNA gene amplicons) and their linked viruses (via vOTU coverages) over time. Viruses were linked to hosts *in silico* by VirMatcher (see Methods). **(B)** Relative abundance-based virus/host ratios of the three most dominant phyla over time.

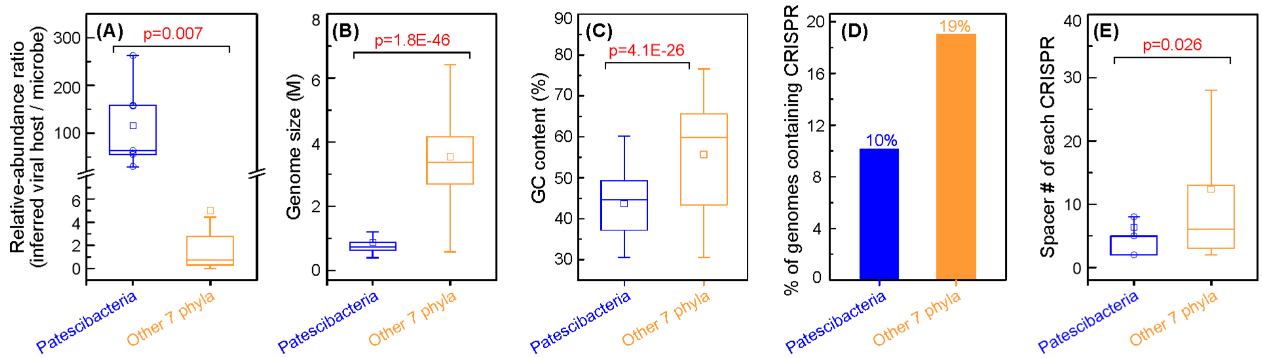
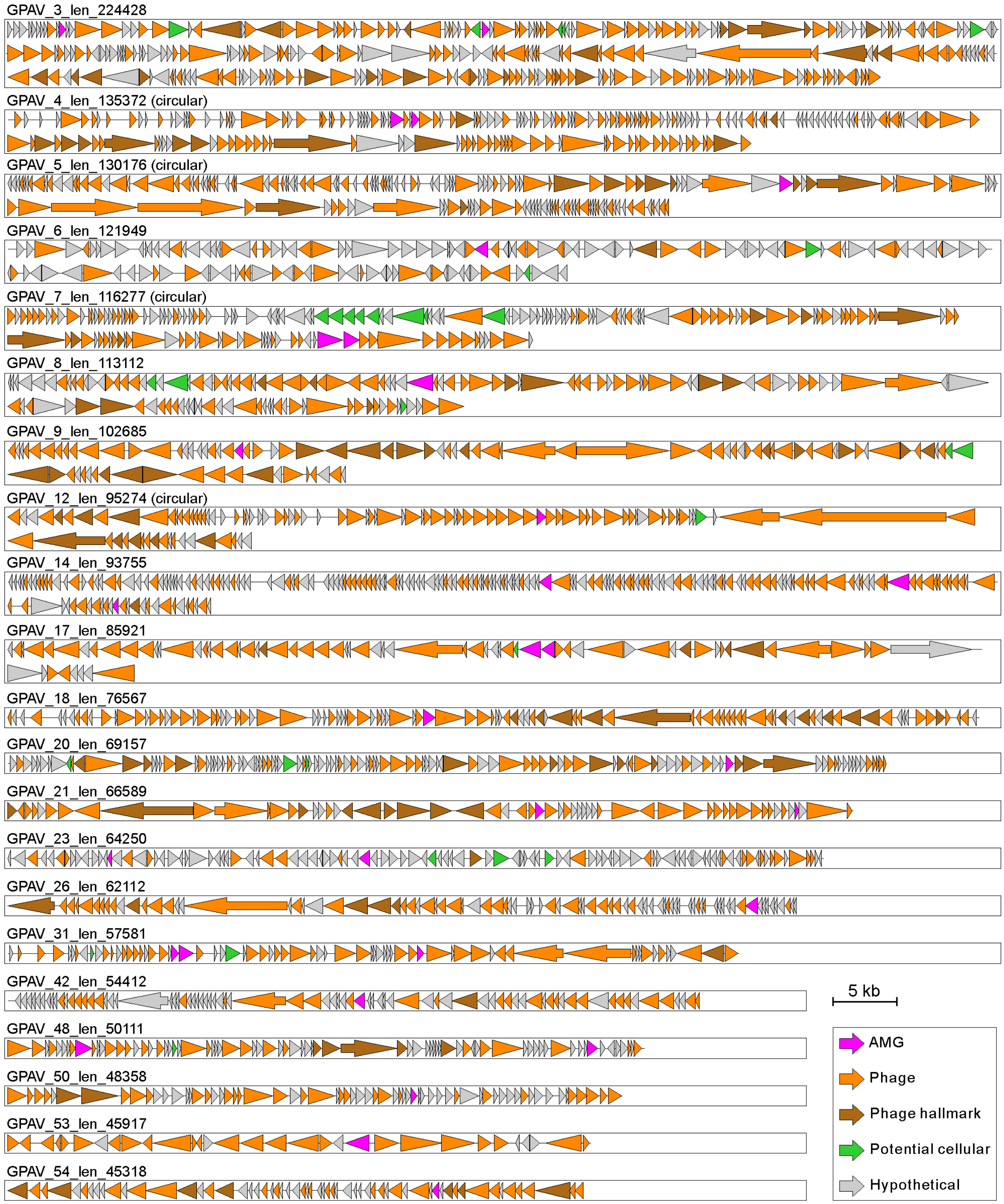


Figure S4. Virus-host abundance ratios and genomic features of Patescibacteria and seven other phyla. Patescibacteria and the seven other phyla (Proteobacteria, Actinobacteria, Bacteroidetes, Patescibacteria, Cyanobacteria, Chloroflexi, Firmicutes, and Verrucomicrobia) represent the top eight abundant phyla in Guliya microbial communities. (**A**) Virus-host abundance ratio (sample size n = 8 and 63 for Patescibacteria and Other 7 phyla, respectively); (**B**) Genome size of microbial MAGs (n = 79 and 3,162); (**C**) GC content (%) of microbial MAGs (n = 79 and 3,162); (**D**) Percentage of microbial MAGs containing ≥1 CRISPR; (**E**) Spacer number of each CRISPR in the microbial MAGs (n = 9 and 962). The p values (two-sided) for comparisons in (A), (B), (C), and (E) were 0.007, 1.8E-46, 4.1E-26, and 0.026, respectively, as indicated by red text in each plot. The 2,350 TP glacier microbial MAGs[7] were used for the genomic feature analyses. Box-plot elements: centre line, median; square symbol, mean; box range, upper and lower quartiles; whiskers, data range of non-outliers; circle (in the left plots of (A) and (E) with n ≤ 10), individual data point. Source data is provided in Supplementary Data 14 and 15.


GPAV_3_len_224428
GPAV_4_len_135372 (circular)
GPAV_5_len_130176 (circular)
GPAV_6_len_121949
GPAV_7_len_116277 (circular)
GPAV_8_len_113112
GPAV_9_len_102685
GPAV_12_len_95274 (circular)
GPAV_14_len_93755
GPAV_17_len_85921
GPAV_18_len_76567
GPAV_20_len_69157
GPAV_21_len_66589
GPAV_23_len_64250
GPAV_26_len_62112
GPAV_31_len_57581
GPAV_42_len_54412
5 kb
GPAV_48_len_50111
AMG
Phage
GPAV_50_len_48358
Phage hallmark
GPAV_53_len_45917
Potential cellular
GPAV_54_len_45318
Hypothetical

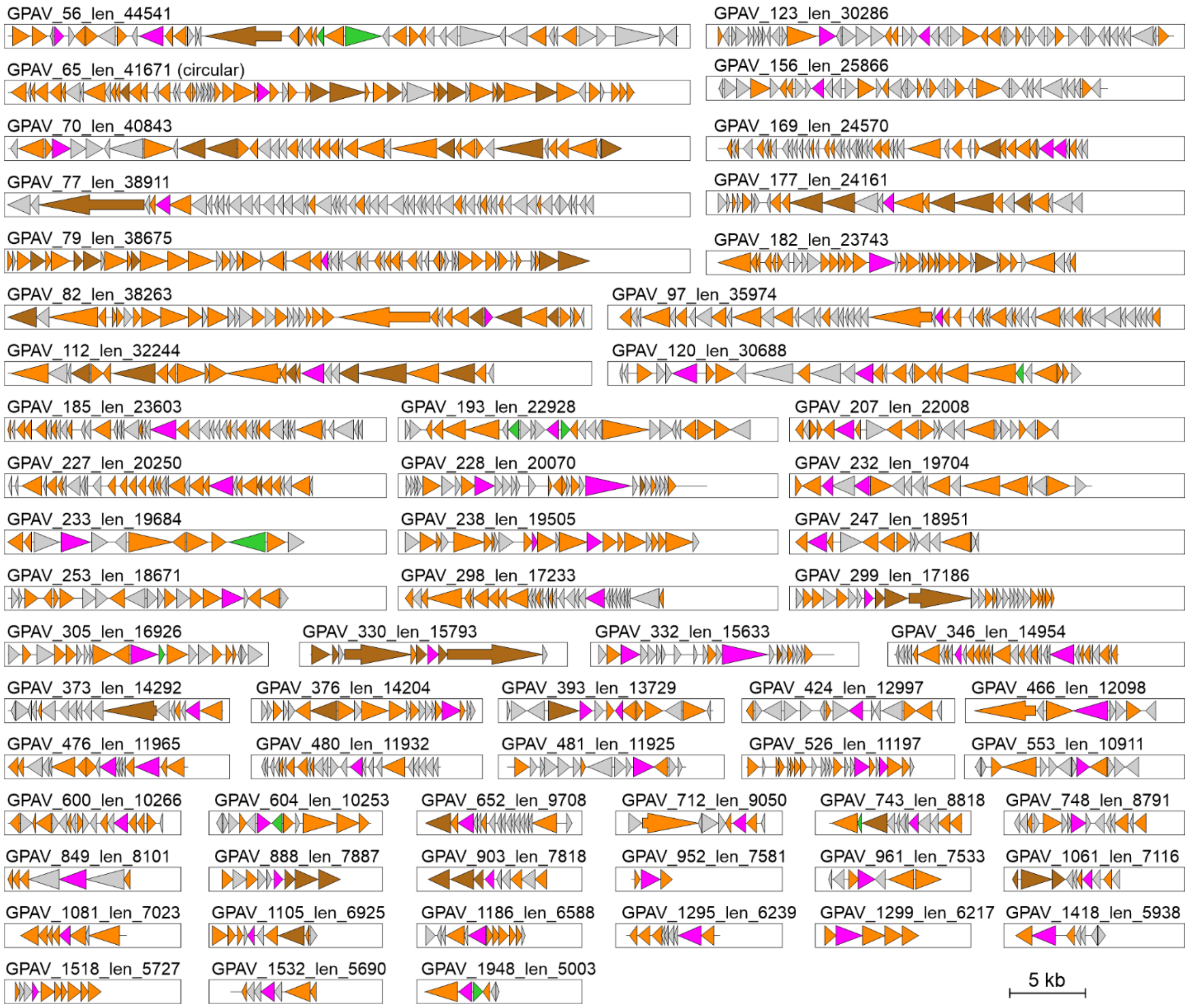
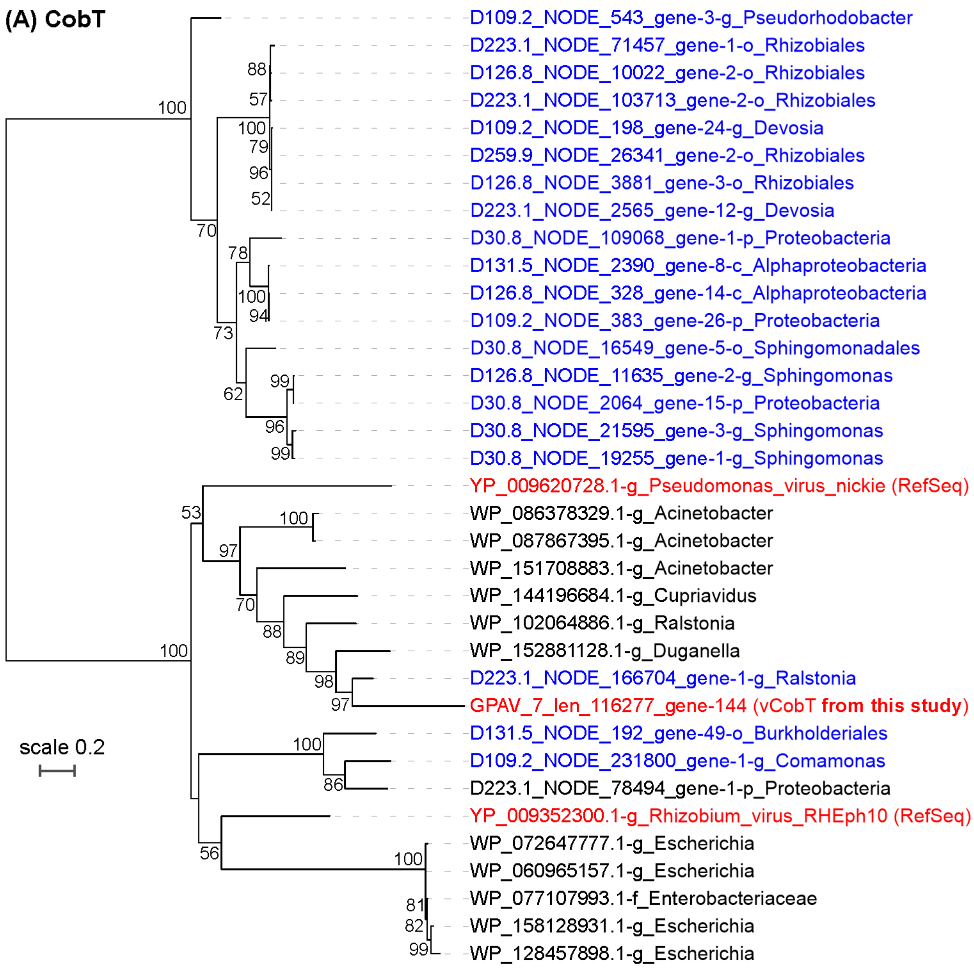


Figure S5. Genome maps of 82 viral contigs containing AMGs. Circular genomes (n=4) are indicated after the contig name. CheckV was used to assess host-virus boundaries and remove potential host fractions on the viral contig. Genes were marked by five colors to illustrate AMGs (purple), phage genes (orange), phage hallmark genes (dark orange), potential cellular genes (green), and hypothetical protein genes (grey). AMGs were detected by VIBRANT and subsequent manual inspections; the latter three groups of genes were classified by comparing their predicted protein sequences to those in CheckV's and VirSorter's databases. Genes were also annotated by comparing them to KEGG and PFAM databases. Genes were marked as "phage genes" if they were matched to the genes of viruses in the VirSorter's or CheckV's virus databases. The putative cellular genes were identified by CheckV and subsequent manual inspections. Genes were considered "hypothetical" if they had no hit to a sequence in any tested databases.


(A) CobT
D109.2_NODE_543_gene-3-g_Pseudorhodobacter
D223.1_NODE_71457_gene-1-o_Rhizobiales
D126.8_NODE_10022_gene-2-o_Rhizobiales
D223.1_NODE_103713_gene-2-o_Rhizobiales
D109.2_NODE_198_gene-24-g_Devosia
D259.9_NODE_26341_gene-2-o_Rhizobiales
D126.8_NODE_3881_gene-3-o_Rhizobiales
D223.1_NODE_2565_gene-12-g_Devosia
D30.8_NODE_109068_gene-1-p_Proteobacteria
D131.5_NODE_2390_gene-8-c_Alphaproteobacteria
D126.8_NODE_328_gene-14-c_Alphaproteobacteria
D109.2_NODE_383_gene-26-p_Proteobacteria
D30.8_NODE_16549_gene-5-o_Sphingomonadales
D126.8_NODE_11635_gene-2-g_Sphingomonas
D30.8_NODE_2064_gene-15-p_Proteobacteria
D30.8_NODE_21595_gene-3-g_Sphingomonas
D30.8_NODE_19255_gene-1-g_Sphingomonas
YP_009620728.1-g_Pseudomonas_virus_nickie (RefSeq)
WP_086378329.1-g_Acinetobacter
WP_087867395.1-g_Acinetobacter
WP_151708883.1-g_Acinetobacter
WP_144196684.1-g_Cupriavidus
WP_102064886.1-g_Ralstonia
WP_152881128.1-g_Duganella
D223.1_NODE_166704_gene-1-g_Ralstonia
GPAV_7_len_116277_gene-144 (vCobT from this study)
D131.5_NODE_192_gene-49-o_Burkholderiales
D109.2_NODE_231800_gene-1-g_Comamonas
D223.1_NODE_78494_gene-1-p_Proteobacteria
YP_009352300.1-g_Rhizobium_virus_RHEph10 (RefSeq)
WP_072647777.1-g_Escherichia
WP_060965157.1-g_Escherichia
WP_077107993.1-f_Enterobacteriaceae
WP_158128931.1-g_Escherichia
WP_128457898.1-g_Escherichia
scale 0.2

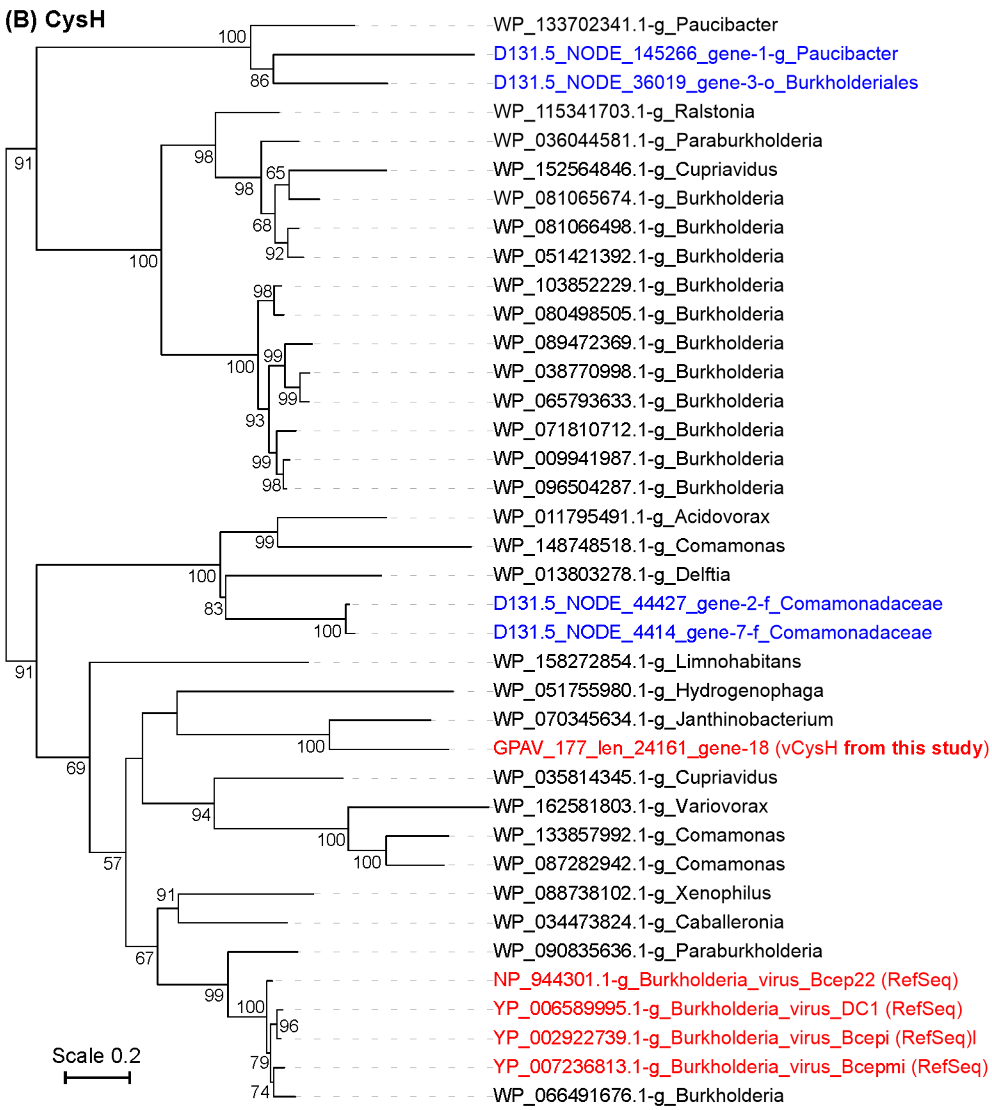
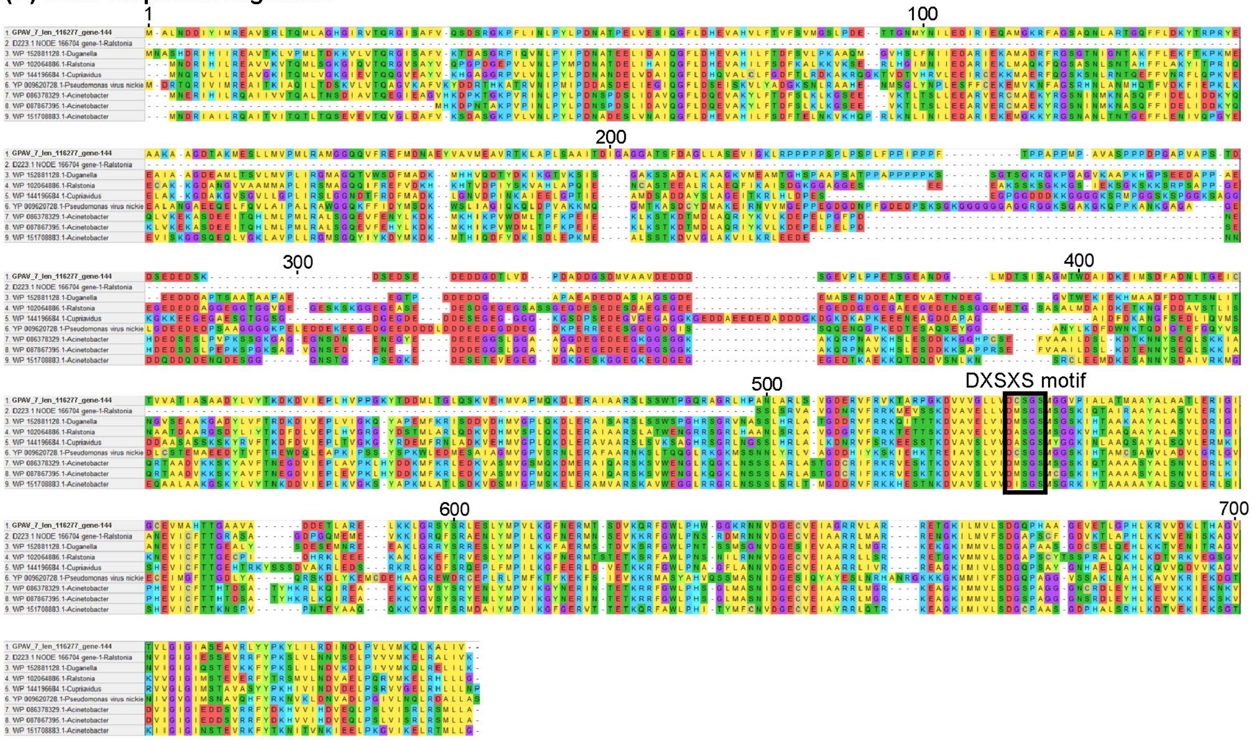


Figure S6. Phylogenetic trees of viral and microbial c*obT* (A) and *cysH* (B) genes. The trees were inferred using maximum-likelihood method with protein sequences (see Methods). Parametric bootstrap values (expressed as percentages of 1,000 replications) ≥40 are shown at branching points. The scale bar indicates a distance of 0.2 substitutions per position in the alignment. The vCobT and vCysH sequences are indicated in red. The microbial sequences (both mCobT and mCysH) obtained from the metagenomes of this study and NCBI RefSeq database are indicated in blue and black, respectively.

(A) CobT sequence alignment



(B) CysH sequence alignment

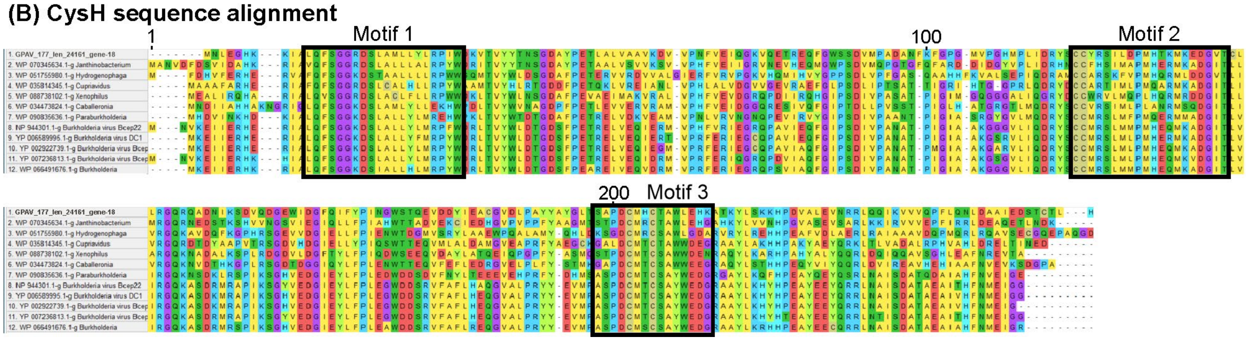


Figure S7. Multiple alignments of viral and mcirobial CobT (A) and CysH (B) protein sequences. The protein sequences of the AMGs and their most closely related bacteria-originated genes were included to illustrate the alignments and conserved motifs. The protein sequences were aligned using MAFFT (v.7.017) with the E-INS-I strategy for 1,000 iterations. The position numbers of aligned sequences are indicated at the top. Conserved motifs were indicated in black boxes. In the conserved motif "DXSXS" of CobT sequences, "X" indicates any amino acid. Putative conserved motifs of CysH squences were identified by the tool MEME[26] (via default parameters) using the alignment of all protein sequences in phylogenetic analysis of Supplementary Fig. S6B.

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